

Zafar Khan

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Professional Experience

Cboe Global Markets, Full Stack Software Engineer

December 2025 - May 2026

- Built a greenfield real-time market data ops dashboard processing millions of daily options events, developing ASP.NET Core Web APIs with Entity Framework Core and PostgreSQL on the backend, React and TypeScript on the frontend, deployed as microservices via Docker and Kubernetes.
- Developed Jenkins CI/CD pipelines and integrated with existing distributed options alert services, delivering new microservices within two-week Agile sprints.
- Built an AI-powered natural language database query system via Model Context Protocol, enabling non-technical production teams to query live PostgreSQL data using Claude and C# without direct database access.
- Developed an AI-driven verification process within Claude Code that automates REST API best practice checks, integrated into the team's C# and TypeScript engineering workflow.
- Designed and implemented an RBAC system supporting 500 users with granular permission controls across the platform.

Wabtec Corporation, Full Stack Software Engineer

June 2022 - December 2025

- Designed and deployed an automated license management system for a global manufacturing enterprise, reducing license provisioning from 30 minutes to under 5 minutes on \$100K+ deals using C#, Python, MySQL, and AWS.
- Delivered a real time remote display system streaming live operational data for external enterprise customers, utilizing WebSockets and OpenSSL for bidirectional server-client communication containerized for deployment (C++, docker)
- Implemented an automated log monitoring pipeline using a machine learning model to detect known defects before customer reports, achieving 99% accuracy (Python, PyTorch)

Wabtec Corporation, Software Engineering Intern/Co-op

May 2021- May 2022

- Created a custom UI for engineers to simulate train runs (playback tool) and included functionality for accurate data to be read. (C# / XAML)
- Automated setup processes, optimizing log parsing and tool runtime efficiency to make the tool over 20x faster. (C#)

The University of Iowa, Undergraduate Teaching Assistant

January 2020 - May 2022

- Guided students in gaining an understanding of the basics of coding (C++)
- Graded homeworks, held office hours, proctored exams, and aided a class of over 300 students learning to code

Education

Penn State University

December 2024

Master of Science in Software Engineering

GPA: 3.8

The University of Iowa

May 2022

Bachelor of Science in Computer Science Engineering

GPA: 3.7 (Presidents List, Old Gold Scholar)

Activities/Projects

The University of Iowa SpaceX and Hyperloop Team, Software Lead

September 2018 - May 2022

- Worked in specific software group to develop and code certain aspects of hyperloop pod (C++)

National Voting System Application (Hackathon top 5 winner)

October 2020

- Developed a full-stack national voting app, featuring custom ballots, real-time voting, and result tracking. (Java / SQL)

Skills

C# • C++ • Python • TypeScript (HTML, CSS) • JavaScript • SQL • .NET • React • EF Core • Git
PostgreSQL • Jenkins • Agile • REST API Design • Kubernetes • PyTorch • LLM • Claude • Docker